

HDOC Day-22

Menu-Driven C Program Using switch

Problem Statement

Write a menu-driven C program that accepts two positive integers m and n , where $m \leq n$. Display a menu and use a switch statement to perform the operation selected by the user. Continue displaying the menu until the user selects Exit, and display an appropriate error message for an invalid choice.

1. Print the multiplication tables of m and n side by side, using multipliers from 1 to 10.
2. Display a multiplication grid containing the products of integers from 1 to m and integers from 1 to n .
3. For each integer from m to n , print that integer followed by the next five consecutive integers.
4. Exit the program.

Algorithm

1. Start.
2. Read two positive integers m and n .
3. Check that $m \leq n$. If not, display an error message and stop.
4. Display the four menu options.
5. Read the user choice.
6. Use a switch statement: Choice 1 prints the tables of m and n from 1 to 10; Choice 2 prints the $m \times n$ multiplication grid; Choice 3 prints each integer from m to n followed by its next five consecutive integers; Choice 4 exits; default displays an invalid-choice message.
7. Repeat the menu until Choice 4 is selected.
8. Stop.

Test-Case Table

Test	m	n	Choice	Expected Result
1	3	5	1	Tables of 3 and 5 from x1 to x10 are displayed side by side.
2	3	5	2	A 3 x 5 multiplication grid is displayed.
3	3	5	3	Rows for 3, 4 and 5 are displayed, each followed by the next five integers.
4	3	5	4	Program exits.
5	3	5	7	Invalid menu choice message is displayed and menu appears again.

C Program

```
#include <stdio.h>
```

```

int main()
{
    int m, n, choice, i, j;

    printf("Enter two positive integers m and n (m <= n): ");
    scanf("%d %d", &m, &n);

    if (m <= 0 || n <= 0 || m > n)
    {
        printf("Invalid input. Enter positive integers with m <= n.\n");
        return 0;
    }

    do
    {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("\nMultiplication Tables of %d and %d\n", m, n);
                for (i = 1; i <= 10; i++)
                {
                    printf("%d x %d = %d\t\t%d x %d = %d\n",
                        m, i, m * i, n, i, n * i);
                }
                break;

            case 2:
                printf("\nMultiplication Grid\n\t");
                for (j = 1; j <= n; j++)
                    printf("%d\t", j);
                printf("\n");

                for (i = 1; i <= m; i++)
                {
                    printf("%d |\t", i);
                    for (j = 1; j <= n; j++)
                        printf("%d\t", i * j);
                    printf("\n");
                }
                break;

            case 3:
                printf("\nConsecutive Integers from %d to %d\n", m, n);
                for (i = m; i <= n; i++)
                {
                    printf("%d: ", i);
                    for (j = i; j <= i + 5; j++)
                        printf("%d ", j);
                    printf("\n");
                }
                break;

            case 4:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice. Please enter 1 to 4.\n");
        }
    } while (choice != 4);

    return 0;
}

```

Compilation

```
(kali~kali) - [~/hdoc22]  
$ vi day22.c  
  
(kali~kali) - [~/hdoc22]  
$ gcc day22.c -o day22
```

Execution 1

```
(kali~kali) - [~/hdoc22]
└─$ ./day22
Enter two positive integers m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 3 and 5
3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50
```

Execution 2

```
(kali@kali) - [~/hdoc22]
$ ./day22
Enter two positive integers m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2

Multiplication Grid
      1  2  3  4  5
-----
1 |  1  2  3  4  5
2 |  2  4  6  8 10
3 |  3  6  9 12 15
```

Execution 3

```
(kali~kali)-[~/hdoc22]
└─$ ./day22
Enter two positive integers m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3

Consecutive Integers from 3 to 5
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10
```

Execution 4

```
(kali~kali)-[~/hdoc22]  
└─$ ./day22  
Enter two positive integers m and n (m <= n): 3 5  
  
1. Multiplication tables of m and n  
2. Multiplication grid  
3. Consecutive integers  
4. Exit  
Enter your choice: 4  
  
Exiting program...
```